

Class 10 NCERT Maths

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Chapter 8: Introduction to Trigonometry (Core-Basics Edition)

CHAPTER 8: INTRODUCTION TO TRIGONOMETRY

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) RIGHT-ANGLED TRIANGLE

Ek angle 90 degree wala triangle. Trigonometry isi pe based hai.

2) TEEN SIDES (angle theta ke reference se)

HYPOTENUSE = 90 degree ke saamne (sabse lambi).

PERPENDICULAR (Opposite) = theta ke saamne wali.

BASE (Adjacent) = theta ke saath wali (hyp nahi).

3) RATIO / FRACTION

Ratio = upar/neeche (jaise 3/5). Trig ratios bhi bas sides ke fractions hain.

4) RECIPROCAL (ulta)

1/x ko x ka reciprocal kehte. sin ka reciprocal cosec, cos ka sec, tan ka cot.

5) SQUARE NOTATION

$\sin^2(A)$ ka matlab $(\sin A)^2 = (\sin A) \times (\sin A)$.

Ye $\sin(A^2)$ nahi hai! Dhyaan rakh.

6) SQUARE ROOT VALUES

$\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$ (yaad rakh).

7) RATIO YAAD KARNE KA TRICK

"Pandit Badri Prasad / Har Har Bole / Sona Chandi Tole"

Perpendicular/Hypotenuse = sin

Base/Hypotenuse = cos

Perpendicular/Base = tan

NOW: ASLI CHAPTER 8 SHURU

TOPIC 1: SIX TRIGONOMETRIC RATIOS

EXAM ALERT: Ye 6 ratios base hain, ratta maar:

$\sin = \text{Perpendicular} / \text{Hypotenuse} \quad (P/H)$

$\cos = \text{Base} / \text{Hypotenuse} \quad (B/H)$

$\tan = \text{Perpendicular} / \text{Base} \quad (P/B)$

$\text{cosec} = 1/\sin = H/P$

$\sec = 1/\cos = H/B$

$\cot = 1/\tan = B/P$

Useful: $\tan = \sin/\cos$, $\cot = \cos/\sin$.

TOPIC 2: SPECIFIC ANGLES TABLE

EXAM ALERT: Ye table HARD yaad kar - har question mein.

Angle:	0	30	45	60	90
sin	0	1/2	1/sq2	sq3/2	1
cos	1	sq3/2	1/sq2	1/2	0
tan	0	1/sq3	1	sq3	undef

(sq2 = sqrt2, sq3 = sqrt3)

[CORE] Trick: sin ke liye 0,1,2,3,4 likho -> /4 ->
sqrt lo. cos = ulta. tan = sin/cos.

TOPIC 3: TRIGONOMETRIC IDENTITIES

EXAM ALERT: Teen identities - proofs mein bahut use:

$\sin^2(A) + \cos^2(A) = 1$

$1 + \tan^2(A) = \sec^2(A)$

$1 + \cot^2(A) = \text{cosec}^2(A)$

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Identity proof

(Exam favourite)

Prove: $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

sec aur cosec ko sin/cos mein likhne se long hota,
simple approach - ek easy version prove karte hain:

Prove: $(1 - \cos^2 A) \times \operatorname{cosec}^2 A = 1$

$$\begin{aligned}\text{LHS} &= \sin^2 A \times \operatorname{cosec}^2 A \quad [1 - \cos^2 = \sin^2] \\ &= \sin^2 A \times (1/\sin^2 A) \quad [\operatorname{cosec} = 1/\sin] \\ &= 1 = \text{RHS. (Proved)}\end{aligned}$$

Solved Example 2 - Ek ratio diya, baaki nikaalo

(Exam favourite)

$\sin A = 3/5$ hai. $\cos A$ aur $\tan A$ nikaalo.

$$\sin^2 + \cos^2 = 1 \rightarrow \cos^2 = 1 - 9/25 = 16/25$$

$$\cos A = 4/5$$

$$\tan A = \sin/\cos = (3/5)/(4/5) = 3/4.$$

Solved Example 3 - Sides se ratios

Right triangle: Perpendicular=5, Base=12. Hyp aur sin, cos, tan?

$$\text{Hyp} = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

$$\sin = 5/13, \cos = 12/13, \tan = 5/12.$$

Solved Example 4 - Specific angles expression

(Exam favourite)

Value: $\sin 60 \cos 30 + \sin 30 \cos 60$

$$= (\sqrt{3}/2)(\sqrt{3}/2) + (1/2)(1/2)$$

$$= 3/4 + 1/4 = 1.$$

Solved Example 5 - Identity check (numbers)

Verify $\sin^2 30 + \cos^2 30 = 1$.

$$\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} + \frac{3}{4} = 1. \text{ (Sahi)}$$

Solved Example 6 - cos given, find others

$\cos A = \frac{12}{13}$. sin A nikaalo.

$$\sin^2 = 1 - \left(\frac{12}{13}\right)^2 = 1 - \frac{144}{169} = \frac{25}{169}$$

$$\sin A = \frac{5}{13}.$$

Solved Example 7 - Simple angle value

$$\tan 45 + \cos 0 = ?$$

$$= 1 + 1 = 2.$$

Solved Example 8 (EASIEST) - Reciprocal

Agar $\sin A = \frac{1}{2}$, toh cosec A = ?

$$\text{cosec} = \frac{1}{\sin} = \frac{1}{(1/2)} = 2.$$

Q AND A TIME

Q1. CORE: sin, cos, tan ko sides ke fraction mein likho.

Q2. Right triangle: Perpendicular=8, Base=15. Hyp aur sin, cos, tan nikaal.

Q3. Value: $\sin 30 + \cos 60$.

Q4. $\cos A = \frac{4}{5}$ hai. sin A aur tan A nikaal.

Q5. Verify: $\sin^2 45 + \cos^2 45 = 1$.

Q6. $\sec^2 60 - \tan^2 60 = ?$ (identity use kar)

Q7. CORE CHECK: cos A ka reciprocal kya hota hai?

SUMMARY

1. $\sin=P/H$, $\cos=B/H$, $\tan=P/B$; reciprocals cosec, sec, cot.

2. Special angles table (0,30,45,60,90) yaad rakh.

3. Identities:

$$\sin^2 + \cos^2 = 1, 1 + \tan^2 = \sec^2, 1 + \cot^2 = \operatorname{cosec}^2.$$

4. $\tan = \sin/\cos$. $\sin^2 A = (\sin A)^2$.

5. $\sqrt{2}=1.414$, $\sqrt{3}=1.732$.

CORE (RED page) revise: right triangle sides, ratio, reciprocal, $\sin^2$ notation, trick "Pandit Badri Prasad".

Generated by Kiro for Krishna1k

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